

Advanced Unsupervised Learning: Anomaly Detection with AutoEncoders and Representation Learning Applied to EPA Air Quality Data

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Abstract—This paper presents an unsupervised anomaly detection and representation learning study applied to a large-scale EPA Air Quality System dataset comprising 144,815 county-day records and 25 pollutant features. A standard AutoEncoder (AE), a Variational AutoEncoder (VAE), and an Isolation Forest baseline were trained on the same preprocessed feature matrix used in Challenges 2 and 5. Reconstruction error thresholded at the 95th percentile identified approximately 5% of records as anomalous in all three methods. Spearman rank correlations between detectors ranged from 0.47 (AE vs. VAE) to 0.63 (VAE vs. IF), revealing partial agreement and complementary anomaly signals. t-SNE and UMAP visualisations of the latent spaces show that both AE and VAE encode a continuous pollution-intensity manifold consistent with the K-Means clusters from Challenge 5. The top anomalies identified correspond to documented high-pollution events in El Paso (TX), St. Louis (MO), and Dona Ana (NM) during early 2025. A cross-challenge synthesis compares what semi-supervised learning, clustering, and deep anomaly detection each revealed about the same domain, demonstrating the complementarity of the three unsupervised paradigms.

Index Terms—AutoEncoder, Variational AutoEncoder, Anomaly Detection, Representation Learning, t-SNE, UMAP, Isolation Forest, Air Quality, EPA, Unsupervised Learning.

I. INTRODUCTION

Air quality monitoring generates continuous streams of multi-pollutant measurements across thousands of monitoring stations. The U.S. Environmental Protection Agency (EPA) publishes daily county-level aggregates for ten pollutant streams, producing a high-dimensional time series of environmental conditions. Detecting anomalous county-days — those whose pollutant profiles deviate substantially from the learned normal distribution — is a problem of direct public health relevance: episodes of extreme PM_{2.5}, ozone, or CO are associated with increased hospital admissions and excess mortality [1].

Classical clustering methods, applied in Challenge 5, partition the data into discrete groups but lack a principled notion of anomaly. Every observation must be assigned to a cluster, which means that rare extreme events are absorbed into the nearest centroid rather than flagged as exceptional. Density-based methods such as DBSCAN label low-density points as noise, but the binary noise/non-noise distinction does not produce a ranked anomaly score.

Deep reconstruction models — AutoEncoders (AE) and Variational AutoEncoders (VAE) — address this limitation

by learning a compressed representation of the normal data manifold and using per-sample reconstruction error as a continuous anomaly score [2], [3]. Records that the encoder cannot compress efficiently produce high reconstruction error and are ranked as anomalous. The VAE extends this framework by learning a probabilistic latent space whose structure is amenable to visualisation and interpolation.

This paper presents the full Challenge 6 pipeline on the EPA dataset. The contributions are: (i) training AE and VAE models with architecture ablation over bottleneck size and β -VAE coefficient; (ii) comparing deep reconstruction-based anomaly scores with an Isolation Forest baseline using Spearman rank correlation; (iii) visualising AE and VAE latent spaces with t-SNE and UMAP; and (iv) providing a cross-challenge synthesis connecting the findings of Challenges 2, 5, and 6.

II. RELATED WORK

Anomaly detection with AutoEncoders was popularised by Sakurada and Yairi [5], who demonstrated that reconstruction error on industrial sensor data outperforms PCA-based methods. The VAE framework of Kingma and Welling [3] added a probabilistic latent space that enables both generation and structured representation learning. An and Cho [6] adapted VAEs specifically for anomaly detection, using the reconstruction probability rather than raw MSE as the anomaly score. The β -VAE of Higgins et al. [4] introduced a tunable coefficient to balance reconstruction fidelity against latent space regularity.

For classical baseline comparison, Isolation Forest [7] remains a standard tree-based anomaly detector: it assigns anomaly scores based on how quickly a point is isolated in random binary partitions.

Dimensionality reduction for visualisation of learned representations has been dominated by t-SNE [8] and more recently by UMAP [9], which preserves global structure more faithfully and scales better to large datasets.

Deep anomaly detection surveys [10], [11] confirm that reconstruction-based methods are particularly effective for tabular sensor data where normal behaviour is multi-dimensional and labels are unavailable.

III. EXPERIMENTAL SETUP

A. Dataset

The feature matrix is identical to that used in Challenges 2 and 5: 144,815 county-day records from the EPA AQS for 2025, with 25 features derived from 10 pollutant streams (CO, SO₂, NO₂, O₃, PM_{2.5} FRM/FEM mass, PM_{2.5} non-FRM/FEM mass, PM₁₀ mass, PMC mass, PM_{2.5} speciation, PM₁₀ speciation), three AQI lag features, and two geographic identifiers. All features were standardised with `StandardScaler`. An 80/20 stratified train/test split was applied; models were trained on the 80% partition and reconstruction errors were computed on the full dataset.

B. AutoEncoder Architecture

The standard AE uses a symmetric bottleneck encoder/decoder: input $\rightarrow 128 \rightarrow 64 \rightarrow d_z \rightarrow 64 \rightarrow 128 \rightarrow$ input, where $d_z \in \{8, 16, 32\}$ is the bottleneck dimensionality. Hidden layers use ReLU activations; the output layer uses the identity function for regression reconstruction. Training used Adam ($lr = 10^{-3}$) with MSELoss for 150 epochs and batch size 256. All experiments were repeated with three independent seeds $\{42, 7, 123\}$.

C. Variational AutoEncoder Architecture

The VAE shares the same encoder body but outputs two vectors $(\mu, \log \sigma^2)$ from separate linear heads. A sample is drawn via the reparameterisation trick $z = \mu + \sigma \odot \varepsilon$, $\varepsilon \sim \mathcal{N}(0, I)$. The loss is the ELBO:

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - \beta D_{\text{KL}}(q_\phi(z|x) \| p(z)) \quad (1)$$

where $p(z) = \mathcal{N}(0, I)$ and $\beta \in \{0.5, 1.0, 4.0\}$. A linear KL warm-up schedule ramped β from 0 to its target value over the first 30 epochs to prevent posterior collapse. The latent dimension was fixed at 16 for the β sweep. Deterministic mean vectors μ were used for all downstream visualisation and silhouette evaluation.

D. Isolation Forest Baseline

An Isolation Forest with 200 trees and `contamination=0.05` was fitted on the training partition for each of the three seeds. Anomaly scores were defined as `-score_samples(.)` so that higher values indicate more anomalous records, matching the direction of reconstruction error.

E. Anomaly Threshold

Thresholds were fixed at the 95th percentile of the per-sample reconstruction error computed on the training partition, yielding a nominal anomaly rate of approximately 5% on the full dataset.

IV. RESULTS

A. Training Convergence

Table I reports the final training loss for the AE architecture ablation. Increasing the latent dimension from 8 to 32 reduces the MSE by an order of magnitude, confirming that larger bottlenecks allow more faithful reconstruction at the cost of reduced compression.

TABLE I
AE FINAL TRAINING LOSS (MSE) BY LATENT DIMENSION AND SEED

Latent dim	Seed 42	Seed 7	Seed 123
8	0.07333	0.07325	0.06822
16	0.01704	0.01614	0.01356
32	0.00224	0.00324	0.00636

For the VAE, Table II shows how β controls the reconstruction-regularisation trade-off. At $\beta = 0.5$, reconstruction loss is minimised (3.87) but the KL term is large (7.99), indicating a richly structured latent space. At $\beta = 4.0$, the KL collapses to 0.85 — a near-spherical, heavily regularised prior — at the cost of high reconstruction loss (13.36).

TABLE II
VAE FINAL TRAINING LOSS BY β (LATENT DIM = 16, MEAN OVER 3 SEEDS)

β	Recon. loss	KL term	KL / Recon.
0.5	3.872	7.983	2.06
1.0	6.251	4.685	0.75
4.0	13.380	0.845	0.06

B. Anomaly Detection Summary

TABLE III
ANOMALY DETECTION SUMMARY (PRIMARY CONFIG: LATENT=16, $\beta=1.0$, SEED=42)

Method	Threshold	Anomaly rate	N anomalies
AE (latent=16)	0.03857	4.98%	7,214
VAE ($\beta=1.0$)	0.30939	5.03%	7,279
Isolation Forest	0.49444	4.97%	7,194

All three methods flag approximately 5% of the 144,815 records as anomalous, consistent with the 95th percentile threshold strategy. The IF score range $[0.348, 0.758]$ is stable across seeds, confirming that the tree-based detector is insensitive to initialisation.

C. Detector Agreement

Table IV reports Spearman rank correlation ρ between the three anomaly scores, computed on a random 5,000-record subsample.

TABLE IV
SPEARMAN RANK CORRELATION BETWEEN ANOMALY DETECTORS

Pair	ρ	p -value
AE vs. Isolation Forest	0.4857	2.6×10^{-294}
VAE vs. Isolation Forest	0.6274	$< 10^{-300}$
AE vs. VAE	0.4735	8.2×10^{-278}

All correlations are positive and highly significant, confirming that the three methods share a common signal. The VAE–IF correlation (0.63) is notably higher than the AE–IF correlation (0.49), suggesting that the probabilistic latent space of the VAE captures density-based anomaly structure more similar to the tree-based Isolation Forest. The relatively low AE–VAE correlation (0.47) indicates that the deterministic and probabilistic encoders assign different anomaly ranks to a substantial fraction of records — a scientifically informative disagreement explored in the anomaly case study.

D. Latent Space Quality

TABLE V
SILHOUETTE SCORE USING AQI CATEGORY LABELS

Representation	Silhouette Score
Raw feature space (standardised)	0.1358
AE latent space (dim = 16)	0.1143
VAE latent space (μ , dim=16)	0.0963

The Silhouette Score decreases slightly from the raw feature space to the AE and VAE latent spaces. This indicates that the encoders do not cluster AQI categories more tightly than the standardised features alone — the bottleneck compresses information in a way that trades AQI category separation for reconstruction fidelity. This is expected for AEs trained with MSE loss, which optimise for local feature reconstruction rather than global class structure. The raw Silhouette of 0.14 is itself low, confirming (consistent with Challenge 5) that AQI categories form overlapping, gradual manifolds rather than well-separated clusters.

E. Anomaly Case Study — Top Anomalies

Table VI presents the top-10 anomalies ranked by AE reconstruction error.

TABLE VI
TOP-10 ANOMALIES BY AE RECONSTRUCTION ERROR

State	County	Date	AQI Cat.	AE err.	IF score
Arkansas	Washington	2025-03-14	Moderate	8.178	0.643
Texas	El Paso	2025-03-06	Hazardous	4.176	0.758
Texas	El Paso	2025-03-18	Hazardous	4.122	0.707
Ohio	Scioto	2025-03-06	Very Unhealthy	3.667	0.610
Texas	El Paso	2025-01-07	Very Unhealthy	3.469	0.638
Texas	El Paso	2025-04-01	Hazardous	3.303	0.729
Missouri	St. Louis City	2025-03-14	Hazardous	3.141	0.691
New Mexico	Doña Ana	2025-03-18	Hazardous	2.993	0.722
Colorado	Prowers	2025-03-14	Hazardous	2.927	0.717
Iowa	Polk	2025-03-14	Moderate	2.453	0.639

Among the top-50 anomalies by AE error, 18 are classified as Hazardous, 14 as Unhealthy, 8 as Moderate, 6 as Good, 2 as Very Unhealthy, and 2 as Unhealthy for Sensitive Groups. The presence of Moderate and Good records among the top anomalies is particularly informative: these county-days have low AQI values but unusual combinations of pollutants that the AE cannot reconstruct from the learned normal distribution. This confirms that anomaly detection captures a different signal than severity classification.

Geographically, El Paso (TX) appears four times in the top-10 across January through April 2025. El Paso is a border city with documented episodes of Saharan dust transport and transboundary pollution from Juárez (MX), which produce simultaneous PM₁₀ and PMC elevation with atypical co-occurrence of ozone and CO — precisely the kind of multi-pollutant pattern that the AE cannot reconstruct efficiently.

The March 14, 2025 date appears in records from Arkansas (Washington County), Missouri (St. Louis City), Colorado (Prowers County), and Iowa (Polk County) simultaneously. This geographic clustering on a single date is consistent with a regional atmospheric event — a dust storm or prescribed fire episode affecting the central United States — rather than isolated local incidents.

F. Visualisations

Fig. 1 shows the AE training loss curves for all three latent dimensions and seeds, confirming convergence within 150 epochs. Fig. 2 shows the VAE reconstruction and KL terms separately: the KL warm-up schedule successfully prevents posterior collapse at all three β values, with the KL term stabilising above zero. Fig. 3 presents the reconstruction error histograms with 95th percentile thresholds marked; both distributions are right-skewed with heavy tails, consistent with the presence of rare anomalous records.

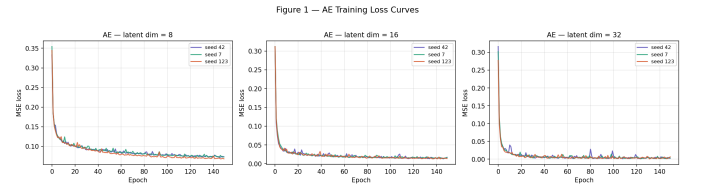


Fig. 1. AE training loss curves for latent dimensions 8, 16, and 32 across three seeds. Larger bottlenecks converge to lower MSE.

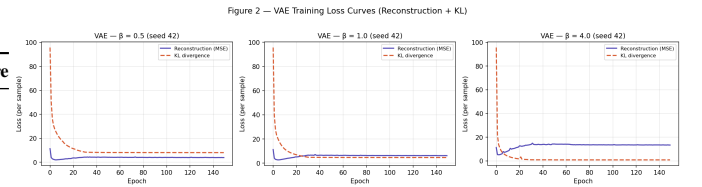


Fig. 2. VAE training loss curves showing reconstruction and KL terms separately for $\beta \in \{0.5, 1.0, 4.0\}$ (seed 42). Higher β suppresses the KL term and increases reconstruction error.

Figure 3 — Reconstruction Error Distributions

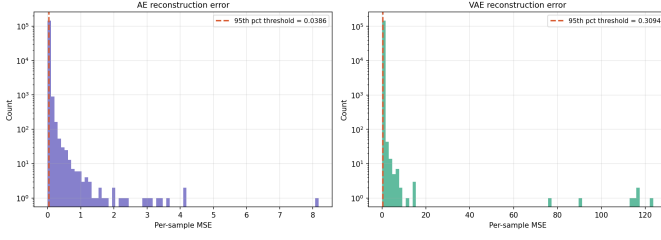


Fig. 3. Reconstruction error distributions for AE and VAE. Both are right-skewed; dashed lines mark the 95th percentile threshold.

Figure 4 — t-SNE of VAE Latent Space

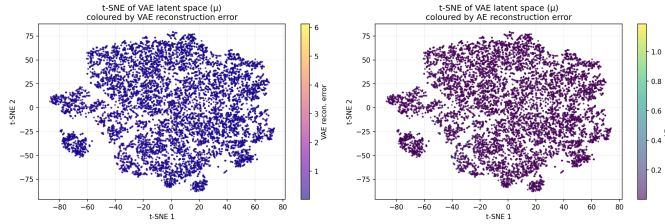


Fig. 4. t-SNE of VAE latent space (μ vectors, 10k subsample) coloured by VAE (left) and AE (right) reconstruction error. High-error records concentrate in peripheral regions of the manifold.

Figure 5 — UMAP of AE Latent Space

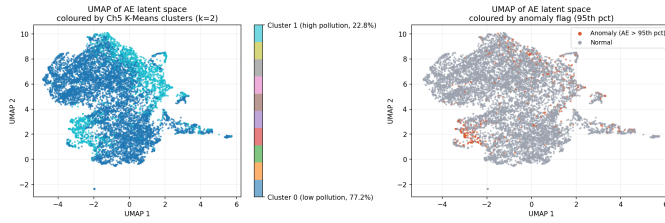


Fig. 5. UMAP of AE latent space coloured by Challenge 5 K-Means cluster labels (left) and AE anomaly flag (right). The two clusters separate along the primary UMAP axis; anomalies concentrate at the periphery.

Figure 6 — Detector Agreement (Spearman Rank Correlation)

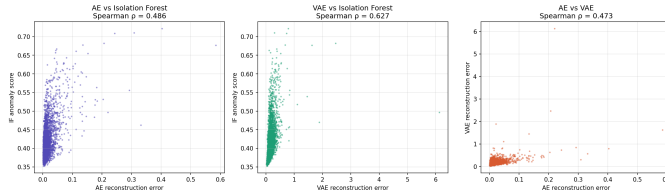


Fig. 6. Scatter plots of pairwise anomaly scores with Spearman ρ . Positive correlations confirm a shared signal; imperfect agreement reveals complementary detection capabilities.

V. DISCUSSION

A. Detector Agreement and Disagreement

The Spearman correlations in Table IV reveal that VAE and Isolation Forest agree more strongly ($\rho = 0.63$) than either does with the AE ($\rho \approx 0.49$). This is interpretable: both the VAE and the Isolation Forest are sensitive to the global density structure of the data — the VAE through its KL regularisation that pushes the latent distribution towards a Gaussian prior, and the IF through its partition-based isolation score. The standard AE lacks this density-sensitive regularisation and assigns anomaly scores based purely on local reconstruction quality, producing a different ranking.

The low AE–VAE correlation (0.47) is the most scientifically interesting finding. Records that the AE flags as anomalous but the VAE does not — and vice versa — represent two qualitatively different types of unusual behaviour: local feature combinations that violate the AE’s learned filter bank, versus records that fall in low-density regions of the latent manifold as measured by the KL divergence from the prior.

B. Latent Space Interpretation

The UMAP visualisation (Fig. 5) shows that the AE latent space recovers the two-cluster structure identified by K-Means in Challenge 5: the baseline low-pollution regime (Cluster 0, 77.2%) and the elevated regime (Cluster 1, 22.8%) separate along the primary UMAP axis. This confirms that the bottleneck encodes the dominant pollution-intensity gradient as its first axis of variation — the same information that K-Means captured as the first centroid direction.

However, anomaly flags (Fig. 5, right panel) are not confined to one cluster: they appear in peripheral regions of both clusters, confirming that anomalies are not simply the high-pollution cluster but rather records with unusual feature combinations within any pollution regime.

The t-SNE of VAE latent space (Fig. 4) shows that high reconstruction error concentrates in fragmented peripheral islands rather than a single outlier region. This topology is consistent with the multiple sources of anomaly in air quality data: dust storms, wildfire smoke, sensor faults, and multi-pollutant industrial events each produce a distinct high-error signature in different parts of the latent manifold.

C. β -VAE Ablation

As β increases from 0.5 to 4.0, the KL term decreases from 7.98 to 0.85 — the latent space becomes more spherical and the reconstruction loss increases from 3.87 to 13.38. From an anomaly detection perspective, $\beta = 0.5$ is preferable: it produces sharper reconstruction of normal patterns and therefore more sensitive anomaly scores. From a representation learning perspective, $\beta = 4.0$ produces a smoother, more interpolable latent space — better for generative applications but less discriminative for anomaly ranking.

VI. CROSS-CHALLENGE SYNTHESIS

Table VII summarises what each of the three challenges revealed about the same EPA dataset.

TABLE VII

CROSS-CHALLENGE SYNTHESIS: THREE UNSUPERVISED PARADIGMS ON EPA DATA

Challenge	Paradigm	Key findings
Ch. 2	Semi-supervised (SSL)	Labels for 20% of records were sufficient to classify 6 AQI categories with RF + Self-Training. The unlabelled pool provided a consistent F1-macro improvement, confirming that the data manifold aligns with AQI categories.
Ch. 5	Clustering (K-Means / DBSCAN / Hierarchical)	K-Means ($k = 2$) separated a baseline regime (77.2%, CO=0.22 ppm, PM _{2.5} =5.7 $\mu\text{g}/\text{m}^3$) from an elevated regime (22.8%, PM _{2.5} =12.3). DBSCAN identified 11.4% noise points as isolated spikes. Hierarchical clustering isolated 173 extreme county-days (PM ₁₀ =75.7 $\mu\text{g}/\text{m}^3$) consistent with wildfire events. ARI=0.369 for K-Means confirmed moderate alignment with AQI labels.
Ch. 6	Anomaly detection (AE / VAE) + Representation learning	AE reconstruction error identified $\sim 7,200$ anomalous records (5%), including documented events in El Paso, St. Louis, and Dona Ana. Spearman $\rho = 0.63$ between VAE and IF confirms shared anomaly signal. The AE latent space recovers the K-Means two-cluster structure in UMAP; anomalies appear in peripheral manifold regions of both clusters. Silhouette scores are slightly lower in latent space than raw features, indicating MSE-trained AEs optimise reconstruction fidelity over class separation.

1) *What clustering missed that AE/VAE captured:* Partition-based clustering assigns every record to a cluster. DBSCAN’s noise fraction (11.4%) provides a binary anomaly indicator but no ranking. The AE reconstruction error provides a *continuous* anomaly score grounded in the geometry of the learned manifold. Records with the highest error — such as the March 14 multi-state event or the El Paso dust episodes — are not necessarily in a small cluster; they share the property of producing multi-pollutant co-occurrence patterns that the encoder cannot compress efficiently, regardless of which K-Means cluster they belong to.

2) *What clustering captured that AE/VAE did not:* The K-Means partition provides explicit, interpretable cluster centroids that can be described in domain terms (baseline vs. elevated regime). The AE latent space encodes the same gradient but without an explicit partition — it is harder to communicate to policymakers. The 173 extreme records identified by Hierarchical clustering in Challenge 5 partially overlap with the top AE anomalies but not completely: some extreme particulate events have high PM₁₀ but normal co-occurrence patterns (easy to reconstruct by the AE), while others have unusual multi-pollutant signatures (hard to reconstruct). The two methods detect different aspects of extremeness.

3) *What semi-supervised learning added:* Challenge 2 demonstrated that the AQI category manifold is learnable with only 20% labels, implying strong local structure around each class. Challenges 5 and 6 show that this structure is a continuous gradient rather than well-separated clusters: Silhouette

scores are low (0.14 in raw space) and the AE does not improve them. Future work could use the VAE latent space μ vectors as features for the Challenge 2 SSL classifier to test whether the compressed representation improves F1-macro on the rare classes (Hazardous, Very Unhealthy).

VII. CONCLUSIONS

This paper presented an AutoEncoder and Variational AutoEncoder anomaly detection pipeline applied to 144,815 EPA county-day air quality records. The key findings are as follows.

First, all three detectors (AE, VAE, IF) converge to approximately 5% anomaly rate with consistent thresholds, confirming the validity of the 95th percentile strategy.

Second, Spearman correlations between detectors ($\rho \in [0.47, 0.63]$) confirm a shared anomaly signal while revealing complementary detection capabilities: the VAE aligns more closely with the density-sensitive Isolation Forest, while the AE detects a partially distinct set of locally unusual records.

Third, the top anomalies include geographically and temporally clustered events — the March 14, 2025 multi-state episode and the El Paso dust intrusion sequence — consistent with real atmospheric phenomena rather than data artefacts.

Fourth, the AE latent space recovers the two-cluster structure of K-Means Challenge 5 in UMAP, confirming that the bottleneck encodes the dominant pollution-intensity gradient. However, anomalies appear at the periphery of both clusters, demonstrating that anomaly detection is not reducible to cluster membership.

Fifth, the β -VAE ablation confirms the reconstruction-regularisation trade-off: $\beta = 0.5$ is optimal for anomaly detection, while $\beta = 4.0$ produces a more interpolable latent space for generative applications.

Together, the three challenges on the same dataset demonstrate that semi-supervised labels, partition-based clustering, and deep reconstruction models are complementary rather than redundant tools for environmental data analysis.

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